Task-5:-Reasoning Document

AWS Services Used in the 3-Tier Deployment

1. Overview

The 3-tier architecture was implemented on AWS to ensure scalability, security, high availability, and ease of management. The architecture consists of:

- **Presentation Layer (Frontend)**
- **Application Layer (Backend)**
- **Data Layer (Database)**
- **CI/CD Layer (Automation)**

Each AWS service was selected based on its suitability, cost-efficiency, and alignment with cloud best practices.

2. AWS Services Used and Justification

2.1 Amazon S3 – Frontend Hosting

Purpose:
Used to host the static frontend application (HTML, CSS, JavaScript).

Why Chosen:

- Ideal for static website hosting
- Highly durable (99.999999999%)
- Cost-effective compared to running a server
- No server management required
- Automatically scalable

2.2 Application Load Balancer – Traffic Distribution

Purpose:
Distributes incoming user requests to the backend EC2 instance.

Why Chosen:

- Performs health checks on backend instances
- Improves availability
- Supports future scaling (Auto Scaling integration)

ALB ensures that traffic is routed only to healthy backend instances, improving reliability.

2.3 Amazon EC2 – Backend Hosting & Jenkins Server

Purpose:

- Hosts the Node.js backend application
- Hosts Jenkins CI/CD server

Why Chosen:

- Full control over server configuration
- Flexible instance sizing
- Suitable for custom backend applications
- Supports installation of additional tools like Jenkins

EC2 provides flexibility and control for application processing and automation tools.

2.4 Amazon RDS – Database Layer

Purpose:
Provides a managed relational database for application data storage.

Why Chosen:

- Automated backups
- Managed patching and maintenance
- Secure private subnet deployment
- High availability options (Multi-AZ)
- Eliminates manual database management

RDS simplifies database operations while ensuring security and reliability.

2.5 Amazon VPC – Network Isolation

Purpose:
Creates a secure virtual network for all resources.

Why Chosen:

- Allows separation of public and private subnets
- Enhances security
- Controls traffic flow using route tables and security groups
- Prevents direct public access to backend and database

2.6 Jenkins – CI/CD Automation

Purpose:

Automates deployment of frontend and backend applications.

Why Chosen:

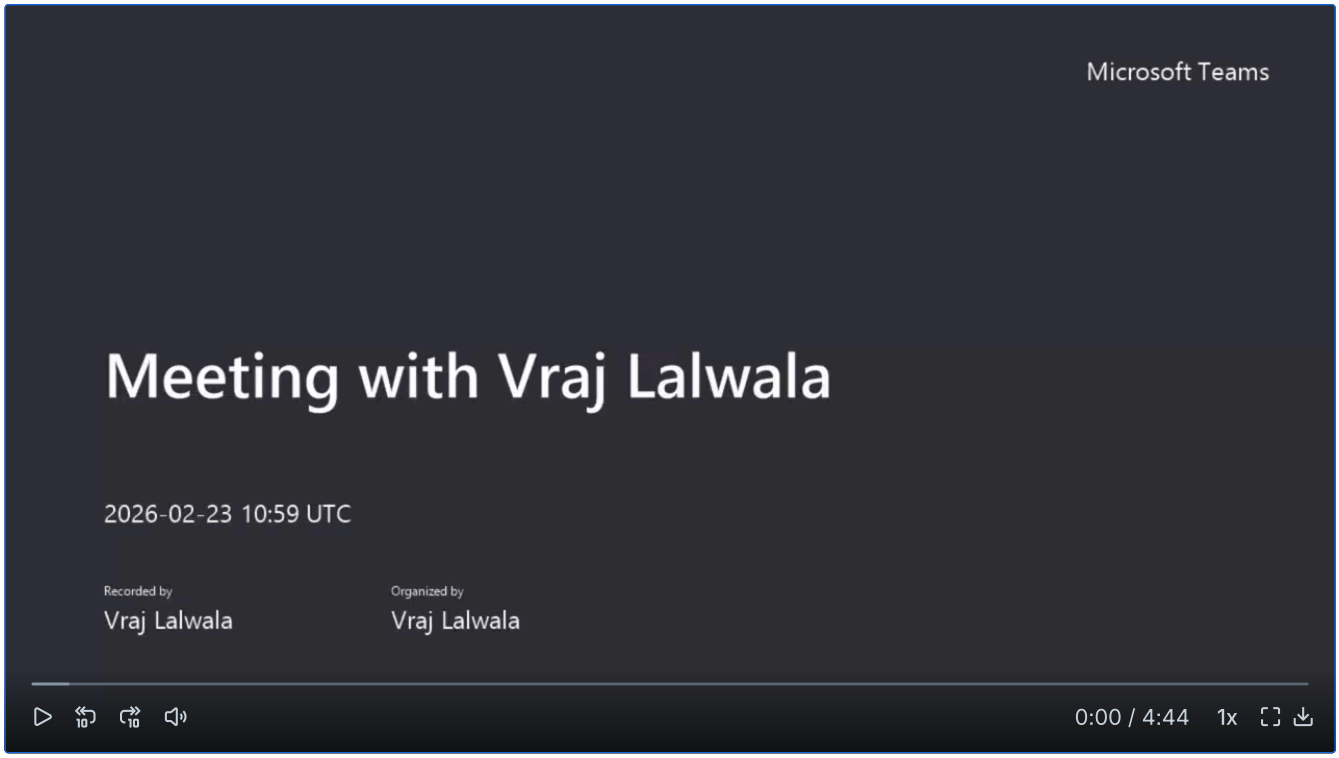
- Open-source and widely used
- Supports integration with GitHub
- Enables automated build and deployment
- Reduces manual errors
- Improves deployment speed

3. Conclusion

The selected AWS services were chosen to build a secure, scalable, and production-ready 3-tier architecture:

- S3 for cost-effective frontend hosting
- ALB for traffic routing and availability
- EC2 for backend processing and CI/CD
- RDS for managed and secure database services
- VPC for network isolation
- Jenkins for automation
- CloudWatch for monitoring

Together, these services provide a robust cloud-native solution that follows AWS best practices for security, scalability, and operational efficiency.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